



## 1. Description

### 1.1. Project

|                 |                    |
|-----------------|--------------------|
| Project Name    | cubeMX             |
| Board Name      | custom             |
| Generated with: | STM32CubeMX 6.16.1 |
| Date            | 05/21/2026         |

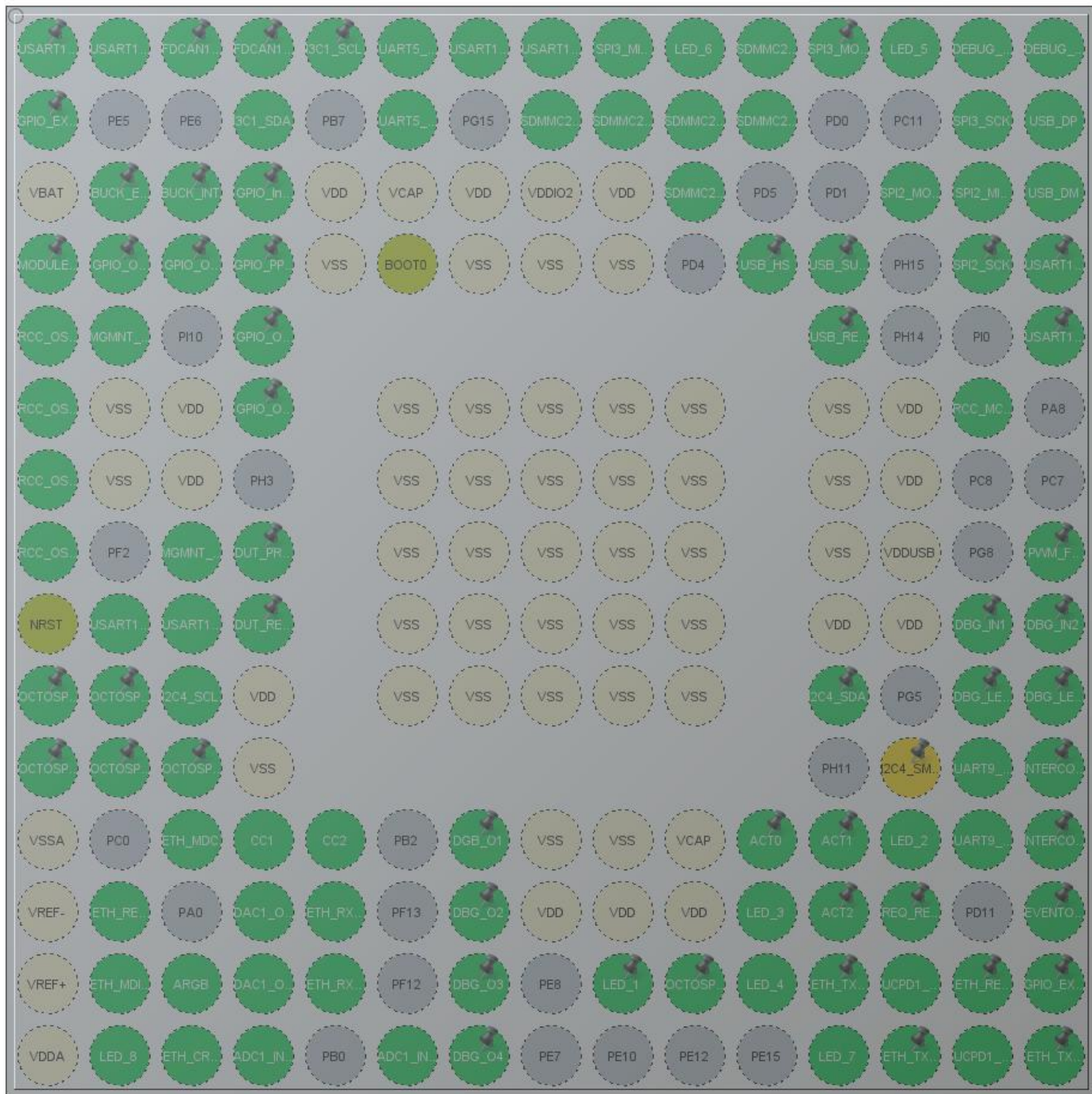
### 1.2. MCU

|                |                |
|----------------|----------------|
| MCU Series     | STM32H5        |
| MCU Line       | STM32H563/H573 |
| MCU name       | STM32H563IIKx  |
| MCU Package    | UFBGA176       |
| MCU Pin number | 201            |

### 1.3. Core(s) information

|         |                |
|---------|----------------|
| Core(s) | ARM Cortex-M33 |
|---------|----------------|

## 2. Pinout Configuration



UFBGA176 +25 (Top view)

### 3. Pins Configuration

| Pin Number<br>UFBGA176 | Pin Name<br>(function after<br>reset) | Pin Type | Alternate<br>Function(s) | Label       |
|------------------------|---------------------------------------|----------|--------------------------|-------------|
| A1                     | PE3                                   | I/O      | USART10_TX               |             |
| A2                     | PE2                                   | I/O      | USART10_RX               |             |
| A3                     | PE1                                   | I/O      | FDCAN1_TX                |             |
| A4                     | PE0                                   | I/O      | FDCAN1_RX                |             |
| A5                     | PB8                                   | I/O      | I3C1_SCL                 |             |
| A6                     | PB5                                   | I/O      | UART5_RX                 |             |
| A7                     | PG14                                  | I/O      | USART10_RTS              |             |
| A8                     | PG13                                  | I/O      | USART10_CTS              |             |
| A9                     | PB4(NJTRST)                           | I/O      | SPI3_MISO                |             |
| A10                    | PB3(JTDO/TRACESWO)                    | I/O      | TIM2_CH2                 | LED_6       |
| A11                    | PD7                                   | I/O      | SDMMC2_CMD               |             |
| A12                    | PC12                                  | I/O      | SPI3_MOSI                |             |
| A13                    | PA15(JTDI)                            | I/O      | TIM2_CH1                 | LED_5       |
| A14                    | PA14(JTCK/SWCLK)                      | I/O      | DEBUG_JTCK-SWCLK         |             |
| A15                    | PA13(JTMS/SWDIO)                      | I/O      | DEBUG_JTMS-SWDIO         |             |
| B1                     | PE4                                   | I/O      | GPIO_EXTI4               |             |
| B4                     | PB9                                   | I/O      | I3C1_SDA                 |             |
| B6                     | PB6                                   | I/O      | UART5_TX                 |             |
| B8                     | PG12                                  | I/O      | SDMMC2_D3                |             |
| B9                     | PG11                                  | I/O      | SDMMC2_D2                |             |
| B10                    | PG10                                  | I/O      | SDMMC2_D1                |             |
| B11                    | PD6                                   | I/O      | SDMMC2_CK                |             |
| B14                    | PC10                                  | I/O      | SPI3_SCK                 |             |
| B15                    | PA12                                  | I/O      | USB_DP                   |             |
| C1                     | VBAT                                  | Power    |                          |             |
| C2                     | PI7 *                                 | I/O      | GPIO_Output              | BUCK_ENABLE |
| C3                     | PI6                                   | I/O      | GPIO_EXTI6               | BUCK_INT    |
| C4                     | PI5 *                                 | I/O      | GPIO_Input               |             |
| C5                     | VDD                                   | Power    |                          |             |
| C6                     | VCAP                                  | Power    |                          |             |
| C7                     | VDD                                   | Power    |                          |             |
| C8                     | VDDIO2                                | Power    |                          |             |
| C9                     | VDD                                   | Power    |                          |             |
| C10                    | PG9                                   | I/O      | SDMMC2_D0                |             |
| C13                    | PI3                                   | I/O      | SPI2_MOSI                |             |
| C14                    | PI2                                   | I/O      | SPI2_MISO                |             |

| Pin Number<br>UFBGA176 | Pin Name<br>(function after<br>reset) | Pin Type | Alternate<br>Function(s) | Label          |
|------------------------|---------------------------------------|----------|--------------------------|----------------|
| C15                    | PA11                                  | I/O      | USB_DM                   |                |
| D1                     | PC13 *                                | I/O      | GPIO_Input               | MODULE_PRESENT |
| D2                     | PI8 *                                 | I/O      | GPIO_Output              | GPIO_OD_1      |
| D3                     | PI9 *                                 | I/O      | GPIO_Output              |                |
| D4                     | PI4 *                                 | I/O      | GPIO_Input               | GPIO_PP_2      |
| D5                     | VSS                                   | Power    |                          |                |
| D6                     | BOOT0                                 | Boot     |                          |                |
| D7                     | VSS                                   | Power    |                          |                |
| D8                     | VSS                                   | Power    |                          |                |
| D9                     | VSS                                   | Power    |                          |                |
| D11                    | PD3 *                                 | I/O      | GPIO_Input               | USB_HS         |
| D12                    | PD2 *                                 | I/O      | GPIO_Input               | USB_SUSP       |
| D14                    | PI1                                   | I/O      | SPI2_SCK                 |                |
| D15                    | PA10                                  | I/O      | USART1_RX                |                |
| E1                     | PC14-<br>OSC32_IN(OSC32_IN)           | I/O      | RCC_OSC32_IN             |                |
| E2                     | PF0                                   | I/O      | I2C2_SDA                 | MGMNT_SDA      |
| E4                     | PI11 *                                | I/O      | GPIO_Input               | GPIO_OD_IN1    |
| E12                    | PH13 *                                | I/O      | GPIO_Output              | USB_RESET      |
| E15                    | PA9                                   | I/O      | USART1_TX                |                |
| F1                     | PC15-<br>OSC32_OUT(OSC32_OUT)         | I/O      | RCC_OSC32_OUT            |                |
| F2                     | VSS                                   | Power    |                          |                |
| F3                     | VDD                                   | Power    |                          |                |
| F4                     | PH2 *                                 | I/O      | GPIO_Input               | GPIO_OD_IN2    |
| F6                     | VSS                                   | Power    |                          |                |
| F7                     | VSS                                   | Power    |                          |                |
| F8                     | VSS                                   | Power    |                          |                |
| F9                     | VSS                                   | Power    |                          |                |
| F10                    | VSS                                   | Power    |                          |                |
| F12                    | VSS                                   | Power    |                          |                |
| F13                    | VDD                                   | Power    |                          |                |
| F14                    | PC9                                   | I/O      | RCC_MCO_2                |                |
| G1                     | PH0-OSC_IN(PH0)                       | I/O      | RCC_OSC_IN               |                |
| G2                     | VSS                                   | Power    |                          |                |
| G3                     | VDD                                   | Power    |                          |                |
| G6                     | VSS                                   | Power    |                          |                |
| G7                     | VSS                                   | Power    |                          |                |
| G8                     | VSS                                   | Power    |                          |                |
| G9                     | VSS                                   | Power    |                          |                |

| Pin Number<br>UFBGA176 | Pin Name<br>(function after<br>reset) | Pin Type | Alternate<br>Function(s) | Label       |
|------------------------|---------------------------------------|----------|--------------------------|-------------|
| G10                    | VSS                                   | Power    |                          |             |
| G12                    | VSS                                   | Power    |                          |             |
| G13                    | VDD                                   | Power    |                          |             |
| H1                     | PH1-OSC_OUT(PH1)                      | I/O      | RCC_OSC_OUT              |             |
| H3                     | PF1                                   | I/O      | I2C2_SCL                 | MGMNT_SCL   |
| H4                     | PH4 *                                 | I/O      | GPIO_Input               | DUT_PRESENT |
| H6                     | VSS                                   | Power    |                          |             |
| H7                     | VSS                                   | Power    |                          |             |
| H8                     | VSS                                   | Power    |                          |             |
| H9                     | VSS                                   | Power    |                          |             |
| H10                    | VSS                                   | Power    |                          |             |
| H12                    | VSS                                   | Power    |                          |             |
| H13                    | VDDUSB                                | Power    |                          |             |
| H15                    | PC6                                   | I/O      | TIM3_CH1                 | PWM_FAN     |
| J1                     | NRST                                  | Reset    |                          |             |
| J2                     | PF3                                   | I/O      | USART11_TX               |             |
| J3                     | PF4                                   | I/O      | USART11_RX               |             |
| J4                     | PH5 *                                 | I/O      | GPIO_Output              | DUT_RESET   |
| J6                     | VSS                                   | Power    |                          |             |
| J7                     | VSS                                   | Power    |                          |             |
| J8                     | VSS                                   | Power    |                          |             |
| J9                     | VSS                                   | Power    |                          |             |
| J10                    | VSS                                   | Power    |                          |             |
| J12                    | VDD                                   | Power    |                          |             |
| J13                    | VDD                                   | Power    |                          |             |
| J14                    | PG7 *                                 | I/O      | GPIO_Input               | DBG_IN1     |
| J15                    | PG6 *                                 | I/O      | GPIO_Input               | DBG_IN2     |
| K1                     | PF7                                   | I/O      | OCTOSPI1_IO2             |             |
| K2                     | PF6                                   | I/O      | OCTOSPI1_IO3             |             |
| K3                     | PF5                                   | I/O      | I2C4_SCL                 |             |
| K4                     | VDD                                   | Power    |                          |             |
| K6                     | VSS                                   | Power    |                          |             |
| K7                     | VSS                                   | Power    |                          |             |
| K8                     | VSS                                   | Power    |                          |             |
| K9                     | VSS                                   | Power    |                          |             |
| K10                    | VSS                                   | Power    |                          |             |
| K12                    | PH12                                  | I/O      | I2C4_SDA                 |             |
| K14                    | PG4 *                                 | I/O      | GPIO_Output              | DBG_LED1    |
| K15                    | PG3 *                                 | I/O      | GPIO_Output              | DBG_LED2    |

| Pin Number<br>UFBGA176 | Pin Name<br>(function after<br>reset) | Pin Type | Alternate<br>Function(s) | Label        |
|------------------------|---------------------------------------|----------|--------------------------|--------------|
| L1                     | PF10                                  | I/O      | OCTOSPI1_CLK             |              |
| L2                     | PF9                                   | I/O      | OCTOSPI1_IO1             |              |
| L3                     | PF8                                   | I/O      | OCTOSPI1_IO0             |              |
| L4                     | VSS                                   | Power    |                          |              |
| L13                    | PH10 **                               | I/O      | I2C4_SMBA                |              |
| L14                    | PD15                                  | I/O      | UART9_TX                 |              |
| L15                    | PG2 *                                 | I/O      | GPIO_Output              | INTERCOM_OUT |
| M1                     | VSSA                                  | Power    |                          |              |
| M3                     | PC1                                   | I/O      | ETH_MDC                  |              |
| M4                     | PC2                                   | I/O      | ADC1_INP12               | CC1          |
| M5                     | PC3                                   | I/O      | ADC1_INP13               | CC2          |
| M7                     | PG1 *                                 | I/O      | GPIO_Output              | DGB_O1       |
| M8                     | VSS                                   | Power    |                          |              |
| M9                     | VSS                                   | Power    |                          |              |
| M10                    | VCAP                                  | Power    |                          |              |
| M11                    | PH6 *                                 | I/O      | GPIO_Input               | ACT0         |
| M12                    | PH8 *                                 | I/O      | GPIO_Input               | ACT1         |
| M13                    | PH9                                   | I/O      | TIM1_CH2                 | LED_2        |
| M14                    | PD14                                  | I/O      | UART9_RX                 |              |
| M15                    | PD13 *                                | I/O      | GPIO_Input               | INTERCOM_IN  |
| N1                     | VREF-                                 | Power    |                          |              |
| N2                     | PA1                                   | I/O      | ETH_REF_CLK              |              |
| N4                     | PA4                                   | I/O      | DAC1_OUT1                |              |
| N5                     | PC4                                   | I/O      | ETH_RXD0                 |              |
| N7                     | PG0 *                                 | I/O      | GPIO_Output              | DBG_O2       |
| N8                     | VDD                                   | Power    |                          |              |
| N9                     | VDD                                   | Power    |                          |              |
| N10                    | VDD                                   | Power    |                          |              |
| N11                    | PE13                                  | I/O      | TIM1_CH3                 | LED_3        |
| N12                    | PH7 *                                 | I/O      | GPIO_Input               | ACT2         |
| N13                    | PD12 *                                | I/O      | GPIO_Input               | REQ_RESET    |
| N15                    | PD10 *                                | I/O      | EVENTOUT                 |              |
| P1                     | VREF+                                 | Power    |                          |              |
| P2                     | PA2                                   | I/O      | ETH_MDIO                 |              |
| P3                     | PA6                                   | I/O      | TIM13_CH1                | ARGB         |
| P4                     | PA5                                   | I/O      | DAC1_OUT2                |              |
| P5                     | PC5                                   | I/O      | ETH_RXD1                 |              |
| P7                     | PF15 *                                | I/O      | GPIO_Output              | DBG_O3       |
| P9                     | PE9                                   | I/O      | TIM1_CH1                 | LED_1        |

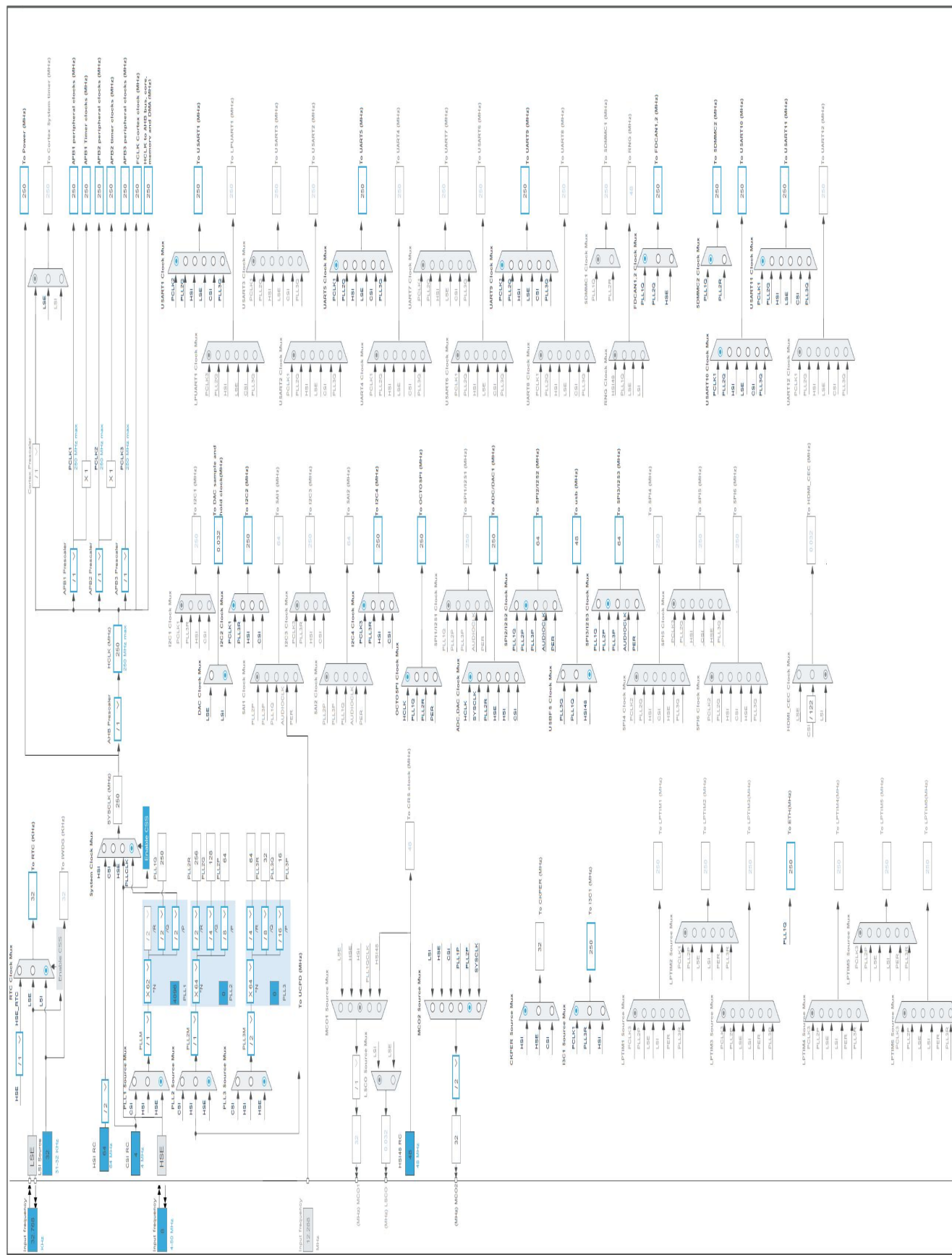
| Pin Number<br>UFBGA176 | Pin Name<br>(function after<br>reset) | Pin Type | Alternate<br>Function(s) | Label     |
|------------------------|---------------------------------------|----------|--------------------------|-----------|
| P10                    | PE11                                  | I/O      | OCTOSPI1_NCS             |           |
| P11                    | PE14                                  | I/O      | TIM1_CH4                 | LED_4     |
| P12                    | PB12                                  | I/O      | ETH_TXD0                 |           |
| P13                    | PB13                                  | I/O      | UCPD1_CC1                |           |
| P14                    | PD9 *                                 | I/O      | GPIO_Output              | ETH_RESET |
| P15                    | PD8                                   | I/O      | GPIO_EXTI8               |           |
| R1                     | VDDA                                  | Power    |                          |           |
| R2                     | PA3                                   | I/O      | TIM2_CH4                 | LED_8     |
| R3                     | PA7                                   | I/O      | ETH_CRS_DV               |           |
| R4                     | PB1                                   | I/O      | ADC1_INP5                |           |
| R6                     | PF11                                  | I/O      | ADC1_INP2                |           |
| R7                     | PF14 *                                | I/O      | GPIO_Output              | DBG_O4    |
| R12                    | PB10                                  | I/O      | TIM2_CH3                 | LED_7     |
| R13                    | PB11                                  | I/O      | ETH_TX_EN                |           |
| R14                    | PB14                                  | I/O      | UCPD1_CC2                |           |
| R15                    | PB15                                  | I/O      | ETH_TXD1                 |           |

\* The pin is affected with an I/O function

\*\* The pin is affected with a peripheral function but no peripheral mode is activated



## 4. Clock Tree Configuration



## 1. Power Consumption Calculator report

### 1.1. Microcontroller Selection

|           |                |
|-----------|----------------|
| Series    | STM32H5        |
| Line      | STM32H563/H573 |
| MCU       | STM32H563IIKx  |
| Datasheet | DS00000 Rev0   |

### 1.2. Parameter Selection

|             |     |
|-------------|-----|
| Temperature | 25  |
| Vdd         | 3.0 |

### 1.3. Battery Selection

|                   |                 |
|-------------------|-----------------|
| Battery           | Li-SOCL2(A3400) |
| Capacity          | 3400.0 mAh      |
| Self Discharge    | 0.08 %/month    |
| Nominal Voltage   | 3.6 V           |
| Max Cont Current  | 100.0 mA        |
| Max Pulse Current | 200.0 mA        |
| Cells in series   | 1               |
| Cells in parallel | 1               |

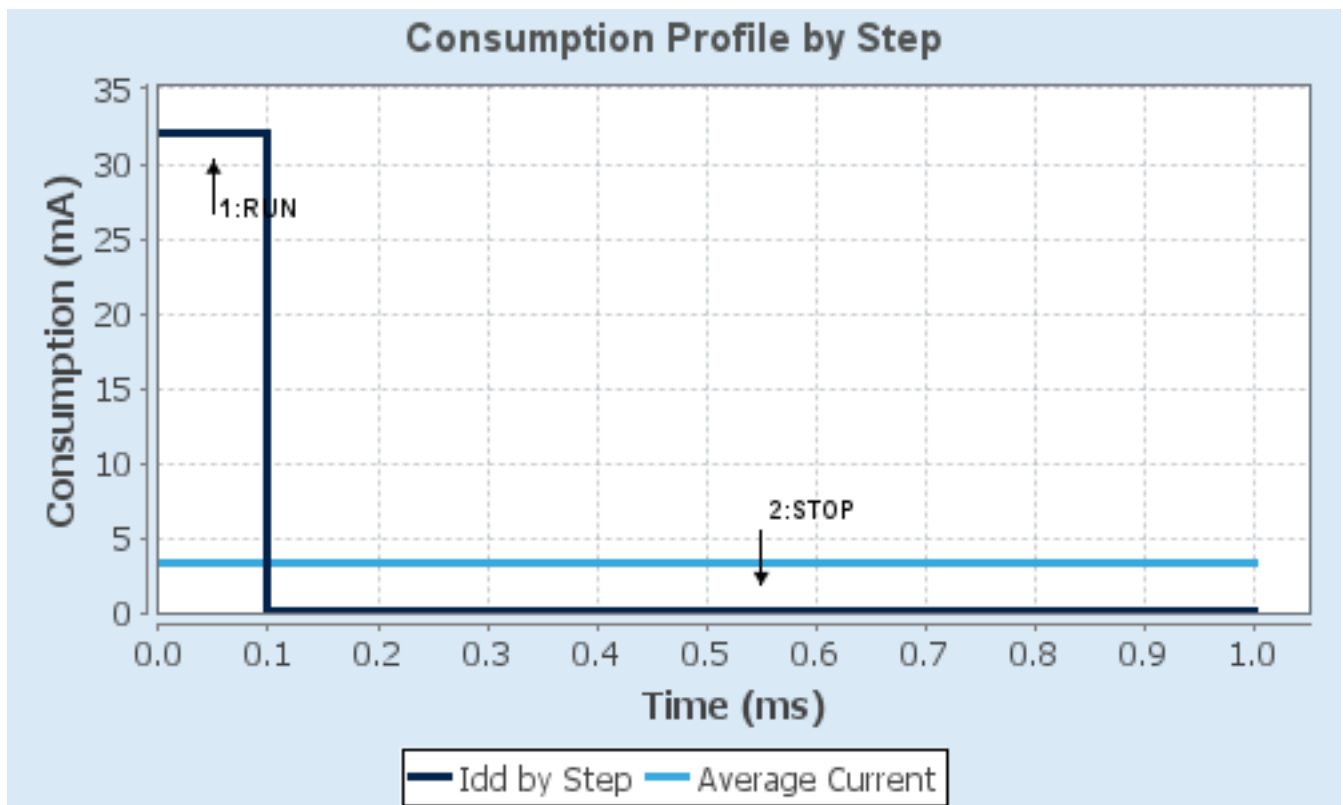
#### 1.4. Sequence

|                               |                                           |                                 |
|-------------------------------|-------------------------------------------|---------------------------------|
| <b>Step</b>                   | Step1                                     | Step2                           |
| <b>Mode</b>                   | RUN                                       | STOP                            |
| <b>Vdd</b>                    | 3.0                                       | 3.0                             |
| <b>Voltage Source</b>         | Battery                                   | Battery                         |
| <b>Range</b>                  | VOS0: Scale0                              | SVOS5: System-Scale5/SMPS       |
| <b>Fetch Type</b>             | FLASH_ON/Cache2Ways_A<br>LL_RAM_RETENTION | Flash-<br>PwrDwn_PwrDwnStop_OFF |
| <b>CPU Frequency</b>          | 250 MHz                                   | 0 Hz                            |
| <b>Clock Configuration</b>    | HSE BYP PLL                               | ALL_CLOCKS_OFF                  |
| <b>Clock Source Frequency</b> | 8 MHz                                     | 0 Hz                            |
| <b>Peripherals</b>            |                                           |                                 |
| <b>Additional Cons.</b>       | 0 mA                                      | 0 mA                            |
| <b>Average Current</b>        | 32 mA                                     | 51.5 $\mu$ A                    |
| <b>Duration</b>               | 0.1 ms                                    | 0.9 ms                          |
| <b>DMIPS</b>                  | 535.0                                     | 0.0                             |
| <b>Ta Max</b>                 | 121.35                                    | 124.99                          |
| <b>Category</b>               | In DS Table                               | In DS Table                     |

#### 1.5. Results

|               |                              |                 |             |
|---------------|------------------------------|-----------------|-------------|
| Sequence Time | 1 ms                         | Average Current | 3.25 mA     |
| Battery Life  | 1 month, 13 days,<br>4 hours | Average DMIPS   | 535.0 DMIPS |

#### 1.6. Chart



## 2. Software Project

### 2.1. Project Settings

| Name                              | Value                         |
|-----------------------------------|-------------------------------|
| Project Name                      | cubeMX                        |
| Project Folder                    | C:\dev\EtherBenchProbe\cubeMX |
| Toolchain / IDE                   | EWARM V9.20                   |
| Firmware Package Name and Version | STM32Cube FW_H5 V1.5.1        |
| Application Structure             | Advanced                      |
| Generate Under Root               | No                            |
| Do not generate the main()        | No                            |
| Minimum Heap Size                 | 0x200                         |
| Minimum Stack Size                | 0x400                         |

### 2.2. Code Generation Settings

| Name                                                            | Value                                           |
|-----------------------------------------------------------------|-------------------------------------------------|
| STM32Cube MCU packages and embedded software                    | Copy all used libraries into the project folder |
| Generate peripheral initialization as a pair of '.c/.h' files   | No                                              |
| Backup previously generated files when re-generating            | No                                              |
| Keep User Code when re-generating                               | Yes                                             |
| Delete previously generated files when not re-generated         | Yes                                             |
| Set all free pins as analog (to optimize the power consumption) | No                                              |
| Enable Full Assert                                              | No                                              |

### 2.3. Advanced Settings - Generated Function Calls

| Rank | Function Name      | Peripheral Instance Name |
|------|--------------------|--------------------------|
| 1    | SystemClock_Config | RCC                      |
| 2    | MX_GPIO_Init       | GPIO                     |
| 3    | MX_ADC1_Init       | ADC1                     |
| 4    | MX_CORDIC_Init     | CORDIC                   |
| 5    | MX_CRC_Init        | CRC                      |
| 6    | MX_DAC1_Init       | DAC1                     |
| 7    | MX_DCACHE1_Init    | DCACHE1                  |
| 8    | MX_ETH_Init        | ETH                      |
| 9    | MX_FDCAN1_Init     | FDCAN1                   |
| 10   | MX_FMAC_Init       | FMAC                     |
| 11   | MX_I2C2_Init       | I2C2                     |

| Rank | Function Name        | Peripheral Instance Name |
|------|----------------------|--------------------------|
| 12   | MX_I2C4_Init         | I2C4                     |
| 13   | MX_I3C1_Init         | I3C1                     |
| 14   | MX_ICACHE_Init       | ICACHE                   |
| 15   | MX_OCTOSPI1_Init     | OCTOSPI1                 |
| 16   | MX_RTC_Init          | RTC                      |
| 17   | MX_SDMMC2_SD_Init    | SDMMC2                   |
| 18   | MX_SPI2_Init         | SPI2                     |
| 19   | MX_SPI3_Init         | SPI3                     |
| 20   | MX_TIM1_Init         | TIM1                     |
| 21   | MX_TIM2_Init         | TIM2                     |
| 22   | MX_TIM3_Init         | TIM3                     |
| 23   | MX_TIM13_Init        | TIM13                    |
| 24   | MX_UART5_Init        | UART5                    |
| 25   | MX_UART9_Init        | UART9                    |
| 26   | MX_USART1_UART_Init  | USART1                   |
| 27   | MX_USART10_UART_Init | USART10                  |
| 28   | MX_USART11_UART_Init | USART11                  |
| 29   | MX_UCPD1_Init        | UCPD1                    |
| 30   | MX_USB_PCD_Init      | USB                      |

## 3. Peripherals and Middlewares Configuration

### 3.1. ADC1

**IN2: IN2 Single-ended**

**IN5: IN5 Single-ended**

**IN12: IN12 Single-ended**

**mode: IN13**

**mode: Temperature Sensor Channel**

**mode: Vrefint Channel**

#### 3.1.1. Parameter Settings:

##### **ADCs\_Common\_Settings:**

Mode Independent mode

##### **ADC\_Settings:**

|                                 |                                                    |
|---------------------------------|----------------------------------------------------|
| Clock Prescaler                 | Asynchronous clock mode divided by 4               |
| Resolution                      | ADC 12-bit resolution                              |
| Scan Conversion Mode            | Disabled                                           |
| Data Alignment                  | Right alignment                                    |
| Continuous Conversion Mode      | Disabled                                           |
| Discontinuous Conversion Mode   | Disabled                                           |
| DMA Continuous Requests         | Disabled                                           |
| End Of Conversion Selection     | End of single conversion                           |
| Overrun behaviour               | Overrun data preserved                             |
| Conversion Data Management Mode | Regular Conversion data stored in DR register only |
| Low Power Auto Wait             | Disabled                                           |

##### **ADC\_Regular\_ConversionMode:**

|                                    |                                         |
|------------------------------------|-----------------------------------------|
| Enable Regular Conversions         | Enable                                  |
| Enable Regular Oversampling        | Disable                                 |
| Number Of Conversion               | 1                                       |
| External Trigger Conversion Source | Regular Conversion launched by software |
| External Trigger Conversion Edge   | None                                    |
| Sampling Mode                      | Normal                                  |
| Rank                               | 1                                       |
| Channel                            | Channel 2                               |
| Sampling Time                      | 2.5 Cycles                              |
| Offset Number                      | No offset                               |
| Monitored by                       | None                                    |

##### **ADC\_Injected\_ConversionMode:**

|                             |         |
|-----------------------------|---------|
| Enable Injected Conversions | Disable |
|-----------------------------|---------|

##### **Analog Watchdog 1:**

|                              |       |
|------------------------------|-------|
| Enable Analog WatchDog1 Mode | false |
|------------------------------|-------|

**Analog Watchdog 2:**

|                              |       |
|------------------------------|-------|
| Enable Analog WatchDog2 Mode | false |
|------------------------------|-------|

**Analog Watchdog 3:**

|                              |       |
|------------------------------|-------|
| Enable Analog WatchDog3 Mode | false |
|------------------------------|-------|

### 3.2. BOOTPATH

**mode: Activated**

### 3.3. CORDIC

**mode: Activated**

### 3.4. CRC

**mode: Activated**

#### 3.4.1. Parameter Settings:

**Basic Parameters:**

|                          |        |
|--------------------------|--------|
| Default Polynomial State | Enable |
|--------------------------|--------|

|                          |        |
|--------------------------|--------|
| Default Init Value State | Enable |
|--------------------------|--------|

**Advanced Parameters:**

|                           |      |
|---------------------------|------|
| Input Data Inversion Mode | None |
|---------------------------|------|

|                            |         |
|----------------------------|---------|
| Output Data Inversion Mode | Disable |
|----------------------------|---------|

|                   |       |
|-------------------|-------|
| Input Data Format | Bytes |
|-------------------|-------|

### 3.5. DAC1

**OUT1 connected to: only external pin**

**OUT2 connected to: only external pin**

#### 3.5.1. Parameter Settings:

**Common DAC Settings:**

|                         |         |
|-------------------------|---------|
| DAC High Frequency Mode | Disable |
|-------------------------|---------|

**DAC Out1 Settings:**

|               |             |
|---------------|-------------|
| Mode selected | Normal Mode |
|---------------|-------------|

|               |        |
|---------------|--------|
| Output Buffer | Enable |
|---------------|--------|

|                      |         |
|----------------------|---------|
| DMA double data mode | Disable |
|----------------------|---------|

|               |         |
|---------------|---------|
| Signed Format | Disable |
|---------------|---------|



|                           |                  |
|---------------------------|------------------|
| Trigger                   | None             |
| User Trimming             | Factory trimming |
| <b>DAC Out2 Settings:</b> |                  |
| Mode selected             | Normal Mode      |
| Output Buffer             | Enable           |
| DMA double data mode      | Disable          |
| Signed Format             | Disable          |
| Trigger                   | None             |
| User Trimming             | Factory trimming |

### 3.6. DCACHE1

**mode: Activated**

#### 3.6.1. Parameter Settings:

**Basic Parameters:**

|                        |                        |
|------------------------|------------------------|
| DCACHE Read Burst Type | DCACHE READ BURST WRAP |
|------------------------|------------------------|

### 3.7. DEBUG

**Debug: Serial Wire**

### 3.8. ETH

**Mode: RMII**

#### 3.8.1. Parameter Settings:

**General : Ethernet Configuration:**

|                      |                   |
|----------------------|-------------------|
| Ethernet MAC Address | 00:80:E1:00:00:00 |
| Tx Descriptor Length | 4                 |
| Rx Descriptor Length | 4                 |
| Rx Buffers Length    | 1524              |

### 3.9. FDCAN1

**mode: Activated**

#### 3.9.1. Parameter Settings:

**Basic Parameters:**

|                         |                          |
|-------------------------|--------------------------|
| Clock Divider           | Divide kernel clock by 1 |
| Frame Format            | Classic mode             |
| Mode                    | Normal mode              |
| Auto Retransmission     | Disable                  |
| Transmit Pause          | Disable                  |
| Protocol Exception      | Disable                  |
| Nominal Sync Jump Width | 1                        |
| Data Prescaler          | 1                        |
| Data Sync Jump Width    | 1                        |
| Data Time Seg1          | 1                        |
| Data Time Seg2          | 1                        |
| Std Filters Nbr         | 0                        |
| Ext Filters Nbr         | 0                        |
| Tx Fifo Queue Mode      | FIFO mode                |

#### Bit Timings Parameters:

|                          |                  |
|--------------------------|------------------|
| Nominal Prescaler        | 16               |
| Nominal Time Quantum     | <b>64.0 *</b>    |
| Nominal Time Seg1        | 1                |
| Nominal Time Seg2        | 1                |
| Nominal Time for one Bit | <b>192 *</b>     |
| Nominal Baud Rate        | <b>5208333 *</b> |

### 3.10. FMAC

**mode: Mode**

### 3.11. I2C2

**I2C: I2C**

#### 3.11.1. Parameter Settings:

##### Timing configuration:

|                               |                     |
|-------------------------------|---------------------|
| I2C Speed Mode                | Standard Mode       |
| I2C Speed Frequency (KHz)     | 100                 |
| Rise Time (ns)                | 0                   |
| Fall Time (ns)                | 0                   |
| Coefficient of Digital Filter | 0                   |
| Analog Filter                 | Enabled             |
| Timing                        | <b>0x60808CD3 *</b> |

##### Slave Features:

|                       |          |
|-----------------------|----------|
| Clock No Stretch Mode | Disabled |
|-----------------------|----------|

|                                  |          |
|----------------------------------|----------|
| General Call Address Detection   | Disabled |
| Primary Address Length selection | 7-bit    |
| Dual Address Acknowledged        | Disabled |
| Primary slave address            | 0        |

### 3.12. I2C4

#### I2C: I2C

##### 3.12.1. Parameter Settings:

###### Timing configuration:

|                               |                     |
|-------------------------------|---------------------|
| I2C Speed Mode                | Standard Mode       |
| I2C Speed Frequency (KHz)     | 100                 |
| Rise Time (ns)                | 0                   |
| Fall Time (ns)                | 0                   |
| Coefficient of Digital Filter | 0                   |
| Analog Filter                 | Enabled             |
| Timing                        | <b>0x60808CD3 *</b> |

###### Slave Features:

|                                  |          |
|----------------------------------|----------|
| Clock No Stretch Mode            | Disabled |
| General Call Address Detection   | Disabled |
| Primary Address Length selection | 7-bit    |
| Dual Address Acknowledged        | Disabled |
| Primary slave address            | 0        |

### 3.13. I3C1

#### Mode: Controller

##### 3.13.1. Parameter Settings:

###### Hardware Specific Timing Constraints:

|                    |     |
|--------------------|-----|
| SDA Rise Time (ns) | 350 |
|--------------------|-----|

###### Bus Characteristics:

|                          |                            |
|--------------------------|----------------------------|
| Bus usage                | <b>Mixed I3C and I2C *</b> |
| Frequency I3C Controller | 1250                       |
| Frequency I2C Controller | 1000                       |
| I3C Duty Cycle           | <b>4 *</b>                 |
| I2C Duty Cycle           | 30                         |
| SCL OD Freq              | <b>1351 *</b>              |

|                                 |                                              |
|---------------------------------|----------------------------------------------|
| SCL PP Freq                     | <b>1250 *</b>                                |
| Timing Register 0               | <b>0x4aae09bd *</b>                          |
| <b>Bus Timing Activity:</b>     |                                              |
| SDA Hold Time                   | SDA Hold 0.5 multiple of Kernel Clock Period |
| Wait Time                       | Own Controller Activity state 0              |
| Bus Free Duration (ns)          | <b>1050 *</b>                                |
| Bus Idle Duration (ns)          | 200000                                       |
| Timing Register 1               | <b>0x008200f8 *</b>                          |
| <b>Basic Configuration:</b>     |                                              |
| Target Hot Join Acknowledgement | Disable                                      |
| Stall Configuration             |                                              |
| - Stall Time (ns)               | 0                                            |
| - ACK Stalling                  | Disable                                      |
| - CCC T-Bit Stalling            | Disable                                      |
| - Tx Parity Stalling            | Disable                                      |
| - Rx Parity Stalling            | Disable                                      |
| Timing Register 2               | 0x00000000                                   |
| <b>Advanced Configuration:</b>  |                                              |
| Own Dynamic Address             | 0                                            |
| SDA High Keeper                 | Disable                                      |
| Fifo Configuration              |                                              |
| - Transmit Fifo Threshold       | Threshold 1 byte                             |
| - Receive Fifo Threshold        | Threshold 1 byte                             |
| - Control Fifo                  | Disable                                      |
| - Status Fifo                   | Disable                                      |

### 3.14. ICACHE

**mode: Memory address remap**

**Mode: 2-ways set associative cache**

#### 3.14.1. Parameter Settings:

##### **Region 0:**

Region Disable

##### **Region 1:**

Region Disable

##### **Region 2:**

Region Disable

##### **Region 3:**

|        |         |
|--------|---------|
| Region | Disable |
|--------|---------|

### 3.15. MEMORYMAP

**mode: Activated**

### 3.16. OCTOSPI1

**Mode: Quad SPI**

**Data: IO[3:0]**

#### 3.16.1. Parameter Settings:

**Generic:**

|                          |                  |
|--------------------------|------------------|
| Fifo Threshold           | 1                |
| Memory Mode              | Single           |
| Memory Type              | Micron           |
| Memory Size              | 16 bits (2 Byte) |
| Chip Select High Time    | 1                |
| Free Running Clock       | Disable          |
| Clock Mode               | Low              |
| Wrap Size                | Not Supported    |
| Clock Prescaler          | 0                |
| Sample Shifting          | None             |
| Delay Hold Quarter Cycle | Disable          |
| Chip Select Boundary     | Disable          |
| Refresh Rate             | 0                |

### 3.17. PWR

**mode: Power saving mode**

**mode: Privilege attributes**

#### 3.17.1. PWR Privilege :

**Privilege PWR:**

|               |         |
|---------------|---------|
| PWR Privilege | Disable |
|---------------|---------|

### 3.18. RCC

**High Speed Clock (HSE): Crystal/Ceramic Resonator**

**Low Speed Clock (LSE) : Crystal/Ceramic Resonator**

**mode: Master Clock Output 2**

### 3.18.1. RCC Privilege :

#### **Privilege RCC:**

|                                   |         |
|-----------------------------------|---------|
| Privilege of RCC Non-Secure Items | Disable |
|-----------------------------------|---------|

### 3.18.2. Parameter Settings:

#### **System Parameters:**

|                         |                    |
|-------------------------|--------------------|
| VDD voltage (V)         | 3.3                |
| Flash Latency(WS)       | 5 WS (6 CPU cycle) |
| Flash Programming Delay | 2                  |

#### **RCC Parameters:**

|                                |          |
|--------------------------------|----------|
| HSI Calibration Value          | 64       |
| CSI Calibration Value          | 32       |
| HSE Startup Timeout Value (ms) | 100      |
| LSE Startup Timeout Value (ms) | 5000     |
| TIM Prescaler Selection        | Disabled |

#### **Power Parameters:**

|                               |                                 |
|-------------------------------|---------------------------------|
| Power Regulator Voltage Scale | Power Regulator Voltage Scale 0 |
|-------------------------------|---------------------------------|

#### **PLL1/2/3 Parameters:**

|                            |                      |
|----------------------------|----------------------|
| PLL1 input frequency range | Between 8 and 16 MHz |
| PLL2 input frequency range | Between 8 and 16 MHz |

## **3.19. RTC**

**mode: Activate Clock Source**

**mode: Activate Calendar**

### 3.19.1. Parameter Settings:

#### **General:**

|                               |                                |
|-------------------------------|--------------------------------|
| Hour Format                   | Hourformat 24                  |
| Asynchronous Predivider value | 127                            |
| Synchronous Predivider value  | 255                            |
| Bin Mode                      | Free running BCD calender mode |

#### **Calendar Time:**

|             |                 |
|-------------|-----------------|
| Data Format | BCD data format |
|-------------|-----------------|

|                                            |                      |
|--------------------------------------------|----------------------|
| Hours                                      | 0                    |
| Minutes                                    | 0                    |
| Seconds                                    | 0                    |
| Day Light Saving: value of hour adjustment | Daylightsaving None  |
| Store Operation                            | Storeoperation Reset |

**Calendar Date:**

|          |         |
|----------|---------|
| Week Day | Monday  |
| Month    | January |
| Date     | 1       |
| Year     | 0       |

### 3.19.2. RTC Privilege:

**Privilege RTC:**

|                    |         |
|--------------------|---------|
| RTC full privilege | Disable |
|--------------------|---------|

**Backup register:**

|              |             |
|--------------|-------------|
| Start zone 1 | RTC_BKP_DR0 |
| Start Zone 2 | RTC_BKP_DR0 |
| start zone 3 | RTC_BKP_DR0 |

**Privilege Backup register :**

|                          |               |
|--------------------------|---------------|
| Backup Register PrivZone | Non-privilege |
|--------------------------|---------------|

**Privilege RTC Feature:**

|                    |               |
|--------------------|---------------|
| RTC Initialisation | Non-Privilege |
| RTC Alarm A        | Non-Privilege |
| RTC Alarm B        | Non-Privilege |
| RTC Calibration    | Non-Privilege |
| RTC TimeStamp      | Non-Privilege |
| RTC WakeUpTimer    | Non-Privilege |

## **3.20. SDMMC2**

### **Mode: SD 4 bits Wide bus**

#### 3.20.1. Parameter Settings:

**SDMMC parameters:**

|                                                   |                                       |
|---------------------------------------------------|---------------------------------------|
| Clock transition on which the bit capture is made | Rising transition                     |
| SDMMC Clock output enable when the bus is idle    | Disable the power save for the clock  |
| SDMMC hardware flow control                       | The hardware control flow is disabled |
| SDMMC clock divide factor                         | 0                                     |
| Is external transceiver present ?                 | no                                    |

### 3.21. SPI2

#### Mode: Full-Duplex Slave

##### 3.21.1. Parameter Settings:

###### Basic Parameters:

|              |           |
|--------------|-----------|
| Frame Format | Motorola  |
| Data Size    | 4 Bits    |
| First Bit    | MSB First |

###### Clock Parameters:

|                       |        |
|-----------------------|--------|
| Clock Polarity (CPOL) | Low    |
| Clock Phase (CPHA)    | 1 Edge |

###### CRC Parameters:

|                 |          |
|-----------------|----------|
| CRC Calculation | Disabled |
|-----------------|----------|

###### Advanced Parameters:

|                            |                              |
|----------------------------|------------------------------|
| NSS Signal Type            | Software                     |
| Fifo Threshold             | Fifo Threshold 01 Data       |
| Nss Polarity               | Nss Polarity Low             |
| Master Ss Idleness         | 00 Cycle                     |
| Master Inter Data Idleness | 00 Cycle                     |
| Master Receiver Auto Susp  | Disable                      |
| Master Keep Io State       | Master Keep Io State Disable |
| IO Swap                    | Disabled                     |
| Ready Signal Polarity      | High                         |

### 3.22. SPI3

#### Mode: Full-Duplex Master

##### 3.22.1. Parameter Settings:

###### Basic Parameters:

|              |           |
|--------------|-----------|
| Frame Format | Motorola  |
| Data Size    | 4 Bits    |
| First Bit    | MSB First |

###### Clock Parameters:

|                           |                       |
|---------------------------|-----------------------|
| Prescaler (for Baud Rate) | 2                     |
| Baud Rate                 | <b>32.0 MBits/s *</b> |
| Clock Polarity (CPOL)     | Low                   |



|                             |                              |
|-----------------------------|------------------------------|
| Clock Phase (CPHA)          | 1 Edge                       |
| <b>CRC Parameters:</b>      |                              |
| CRC Calculation             | Disabled                     |
| <b>Advanced Parameters:</b> |                              |
| NSSP Mode                   | Enabled                      |
| NSS Signal Type             | Software                     |
| Fifo Threshold              | Fifo Threshold 01 Data       |
| Nss Polarity                | Nss Polarity Low             |
| Master Ss Idleness          | 00 Cycle                     |
| Master Inter Data Idleness  | 00 Cycle                     |
| Master Receiver Auto Susp   | Disable                      |
| Master Keep Io State        | Master Keep Io State Disable |
| IO Swap                     | Disabled                     |
| Ready Master Management     | Internal                     |
| Ready Signal Polarity       | High                         |

### 3.23. SYS

**Timebase Source: TIM6**

### 3.24. TIM1

**Clock Source : Internal Clock**

**Channel1: PWM Generation CH1**

**Channel2: PWM Generation CH2**

**Channel3: Output Compare CH3**

**Channel4: PWM Generation CH4**

#### 3.24.1. Parameter Settings:

##### **Counter Settings:**

|                                                       |                                            |
|-------------------------------------------------------|--------------------------------------------|
| Prescaler (PSC - 16 bits value)                       | 0                                          |
| Counter Mode                                          | Up                                         |
| Dithering                                             | Disable                                    |
| Counter Period (AutoReload Register - 16 bits value ) | 65535                                      |
| Internal Clock Division (CKD)                         | No Division                                |
| Repetition Counter (RCR - 16 bits value)              | 0                                          |
| auto-reload preload                                   | Disable                                    |
| <b>Trigger Output (TRGO) Parameters:</b>              |                                            |
| Master/Slave Mode (MSM bit)                           | Disable (Trigger input effect not delayed) |
| Trigger Event Selection TRGO                          | Reset (UG bit from TIMx_EGR)               |
| Trigger Event Selection TRGO2                         | Reset (UG bit from TIMx_EGR)               |

#### Break And Dead Time management - BRK Configuration:

|                           |         |
|---------------------------|---------|
| BRK State                 | Disable |
| BRK Polarity              | High    |
| BRK Filter (4 bits value) | 0       |
| BRK Sources Configuration |         |
| - Digital Input           | Disable |

#### Break And Dead Time management - BRK2 Configuration:

|                            |         |
|----------------------------|---------|
| BRK2 State                 | Disable |
| BRK2 Polarity              | High    |
| BRK2 Filter (4 bits value) | 0       |
| BRK2 Sources Configuration |         |
| - Digital Input            | Disable |

#### Break And Dead Time management - Output Configuration:

|                                          |         |
|------------------------------------------|---------|
| Automatic Output State                   | Disable |
| Off State Selection for Run Mode (OSSR)  | Disable |
| Off State Selection for Idle Mode (OSSI) | Disable |
| Lock Configuration                       | Off     |

#### Clear Input:

|                    |         |
|--------------------|---------|
| Clear Input Source | Disable |
|--------------------|---------|

#### PWM Generation Channel 1:

|                        |            |
|------------------------|------------|
| Mode                   | PWM mode 1 |
| Pulse (16 bits value)  | 0          |
| Output compare preload | Enable     |
| Fast Mode              | Disable    |
| CH Polarity            | High       |
| CH Idle State          | Reset      |

#### PWM Generation Channel 2:

|                        |            |
|------------------------|------------|
| Mode                   | PWM mode 1 |
| Pulse (16 bits value)  | 0          |
| Output compare preload | Enable     |
| Fast Mode              | Disable    |
| CH Polarity            | High       |
| CH Idle State          | Reset      |

#### Output Compare Channel 3:

|                        |                               |
|------------------------|-------------------------------|
| Mode                   | Frozen (used for Timing base) |
| Pulse (16 bits value)  | 0                             |
| Output compare preload | Disable                       |
| CH Polarity            | High                          |
| CH Idle State          | Reset                         |

#### PWM Generation Channel 4:

|      |            |
|------|------------|
| Mode | PWM mode 1 |
|------|------------|

|                        |         |
|------------------------|---------|
| Pulse (16 bits value)  | 0       |
| Output compare preload | Enable  |
| Fast Mode              | Disable |
| CH Polarity            | High    |
| CH Idle State          | Reset   |

### 3.25. TIM2

**Clock Source : Internal Clock**

**Channel1: PWM Generation CH1**

**Channel2: PWM Generation CH2**

**Channel3: PWM Generation CH3**

**Channel4: PWM Generation CH4**

#### 3.25.1. Parameter Settings:

##### **Counter Settings:**

|                                                       |             |
|-------------------------------------------------------|-------------|
| Prescaler (PSC - 16 bits value)                       | 0           |
| Counter Mode                                          | Up          |
| Dithering                                             | Disable     |
| Counter Period (AutoReload Register - 32 bits value ) | 4294967295  |
| Internal Clock Division (CKD)                         | No Division |
| auto-reload preload                                   | Disable     |

##### **Trigger Output (TRGO) Parameters:**

|                              |                                            |
|------------------------------|--------------------------------------------|
| Master/Slave Mode (MSM bit)  | Disable (Trigger input effect not delayed) |
| Trigger Event Selection TRGO | Reset (UG bit from TIMx_EGR)               |

##### **Clear Input:**

|                    |         |
|--------------------|---------|
| Clear Input Source | Disable |
|--------------------|---------|

##### **PWM Generation Channel 1:**

|                        |            |
|------------------------|------------|
| Mode                   | PWM mode 1 |
| Pulse (32 bits value)  | 0          |
| Output compare preload | Enable     |
| Fast Mode              | Disable    |
| CH Polarity            | High       |

##### **PWM Generation Channel 2:**

|                        |            |
|------------------------|------------|
| Mode                   | PWM mode 1 |
| Pulse (32 bits value)  | 0          |
| Output compare preload | Enable     |
| Fast Mode              | Disable    |
| CH Polarity            | High       |

##### **PWM Generation Channel 3:**

|                        |            |
|------------------------|------------|
| Mode                   | PWM mode 1 |
| Pulse (32 bits value)  | 0          |
| Output compare preload | Enable     |
| Fast Mode              | Disable    |
| CH Polarity            | High       |

#### **PWM Generation Channel 4:**

|                        |            |
|------------------------|------------|
| Mode                   | PWM mode 1 |
| Pulse (32 bits value)  | 0          |
| Output compare preload | Enable     |
| Fast Mode              | Disable    |
| CH Polarity            | High       |

### **3.26. TIM3**

**Clock Source : Internal Clock**

**Channel1: PWM Generation CH1**

#### 3.26.1. Parameter Settings:

##### **Counter Settings:**

|                                                       |             |
|-------------------------------------------------------|-------------|
| Prescaler (PSC - 16 bits value)                       | 0           |
| Counter Mode                                          | Up          |
| Dithering                                             | Disable     |
| Counter Period (AutoReload Register - 16 bits value ) | 65535       |
| Internal Clock Division (CKD)                         | No Division |
| auto-reload preload                                   | Disable     |

##### **Trigger Output (TRGO) Parameters:**

|                              |                                            |
|------------------------------|--------------------------------------------|
| Master/Slave Mode (MSM bit)  | Disable (Trigger input effect not delayed) |
| Trigger Event Selection TRGO | Reset (UG bit from TIMx_EGR)               |

##### **Clear Input:**

|                    |         |
|--------------------|---------|
| Clear Input Source | Disable |
|--------------------|---------|

#### **PWM Generation Channel 1:**

|                        |            |
|------------------------|------------|
| Mode                   | PWM mode 1 |
| Pulse (16 bits value)  | 0          |
| Output compare preload | Enable     |
| Fast Mode              | Disable    |
| CH Polarity            | High       |

### **3.27. TIM13**

**mode: Activated**

## **Channel1: PWM Generation CH1**

### 3.27.1. Parameter Settings:

#### **Counter Settings:**

|                                                       |             |
|-------------------------------------------------------|-------------|
| Prescaler (PSC - 16 bits value)                       | 0           |
| Counter Mode                                          | Up          |
| Dithering                                             | Disable     |
| Counter Period (AutoReload Register - 16 bits value ) | 65535       |
| Internal Clock Division (CKD)                         | No Division |
| auto-reload preload                                   | Disable     |

#### **Clear Input:**

|                    |         |
|--------------------|---------|
| Clear Input Source | Disable |
|--------------------|---------|

#### **PWM Generation Channel 1:**

|                        |            |
|------------------------|------------|
| Mode                   | PWM mode 1 |
| Pulse (16 bits value)  | 0          |
| Output compare preload | Enable     |
| Fast Mode              | Disable    |
| CH Polarity            | High       |

## **3.28. UART5**

**Mode: Asynchronous**

### 3.28.1. Parameter Settings:

#### **Basic Parameters:**

|             |                           |
|-------------|---------------------------|
| Baud Rate   | 115200                    |
| Word Length | 8 Bits (including Parity) |
| Parity      | None                      |
| Stop Bits   | 1                         |

#### **Advanced Parameters:**

|                  |                             |
|------------------|-----------------------------|
| Data Direction   | Receive and Transmit        |
| Over Sampling    | 16 Samples                  |
| Single Sample    | Disable                     |
| ClockPrescaler   | 1                           |
| Fifo Mode        | FIFO mode disable           |
| Txfifo Threshold | 1 eighth full configuration |
| Rxfifo Threshold | 1 eighth full configuration |

#### **Advanced Features:**

|               |         |
|---------------|---------|
| Auto Baudrate | Disable |
|---------------|---------|

|                               |         |
|-------------------------------|---------|
| TX Pin Active Level Inversion | Disable |
| RX Pin Active Level Inversion | Disable |
| Data Inversion                | Disable |
| TX and RX Pins Swapping       | Disable |
| Overrun                       | Enable  |
| DMA on RX Error               | Enable  |
| MSB First                     | Disable |

### 3.29. UART9

#### Mode: Asynchronous

##### 3.29.1. Parameter Settings:

###### Basic Parameters:

|             |                           |
|-------------|---------------------------|
| Baud Rate   | 115200                    |
| Word Length | 8 Bits (including Parity) |
| Parity      | None                      |
| Stop Bits   | 1                         |

###### Advanced Parameters:

|                  |                             |
|------------------|-----------------------------|
| Data Direction   | Receive and Transmit        |
| Over Sampling    | 16 Samples                  |
| Single Sample    | Disable                     |
| ClockPrescaler   | 1                           |
| Fifo Mode        | FIFO mode disable           |
| Txfifo Threshold | 1 eighth full configuration |
| Rxfifo Threshold | 1 eighth full configuration |

###### Advanced Features:

|                               |         |
|-------------------------------|---------|
| Auto Baudrate                 | Disable |
| TX Pin Active Level Inversion | Disable |
| RX Pin Active Level Inversion | Disable |
| Data Inversion                | Disable |
| TX and RX Pins Swapping       | Disable |
| Overrun                       | Enable  |
| DMA on RX Error               | Enable  |
| MSB First                     | Disable |

### 3.30. UCPD1

#### UCPD Mode: Sink

### 3.30.1. Parameter Settings:

Version 1.0

## **3.31. USART1**

**Mode: Asynchronous**

### 3.31.1. Parameter Settings:

#### **Basic Parameters:**

|             |                           |
|-------------|---------------------------|
| Baud Rate   | 115200                    |
| Word Length | 8 Bits (including Parity) |
| Parity      | None                      |
| Stop Bits   | 1                         |

#### **Advanced Parameters:**

|                  |                             |
|------------------|-----------------------------|
| Data Direction   | Receive and Transmit        |
| Over Sampling    | 16 Samples                  |
| Single Sample    | Disable                     |
| ClockPrescaler   | 1                           |
| Fifo Mode        | Disable                     |
| Txfifo Threshold | 1 eighth full configuration |
| Rxfifo Threshold | 1 eighth full configuration |

#### **Advanced Features:**

|                               |         |
|-------------------------------|---------|
| Auto Baudrate                 | Disable |
| TX Pin Active Level Inversion | Disable |
| RX Pin Active Level Inversion | Disable |
| Data Inversion                | Disable |
| TX and RX Pins Swapping       | Disable |
| Overrun                       | Enable  |
| DMA on RX Error               | Enable  |
| MSB First                     | Disable |

## **3.32. USART10**

**Mode: Asynchronous**

**Hardware Flow Control (RS232): CTS/RTS**

### 3.32.1. Parameter Settings:

#### **Basic Parameters:**

|             |                           |
|-------------|---------------------------|
| Baud Rate   | 115200                    |
| Word Length | 8 Bits (including Parity) |
| Parity      | None                      |
| Stop Bits   | 1                         |

**Advanced Parameters:**

|                  |                             |
|------------------|-----------------------------|
| Data Direction   | Receive and Transmit        |
| Over Sampling    | 16 Samples                  |
| Single Sample    | Disable                     |
| ClockPrescaler   | 1                           |
| Fifo Mode        | Disable                     |
| Txfifo Threshold | 1 eighth full configuration |
| Rxfifo Threshold | 1 eighth full configuration |

**Advanced Features:**

|                               |         |
|-------------------------------|---------|
| Auto Baudrate                 | Disable |
| TX Pin Active Level Inversion | Disable |
| RX Pin Active Level Inversion | Disable |
| Data Inversion                | Disable |
| TX and RX Pins Swapping       | Disable |
| Overrun                       | Enable  |
| DMA on RX Error               | Enable  |
| MSB First                     | Disable |

### 3.33. USART11

#### Mode: Asynchronous

##### 3.33.1. Parameter Settings:

**Basic Parameters:**

|             |                           |
|-------------|---------------------------|
| Baud Rate   | 115200                    |
| Word Length | 8 Bits (including Parity) |
| Parity      | None                      |
| Stop Bits   | 1                         |

**Advanced Parameters:**

|                  |                             |
|------------------|-----------------------------|
| Data Direction   | Receive and Transmit        |
| Over Sampling    | 16 Samples                  |
| Single Sample    | Disable                     |
| ClockPrescaler   | 1                           |
| Fifo Mode        | Disable                     |
| Txfifo Threshold | 1 eighth full configuration |
| Rxfifo Threshold | 1 eighth full configuration |

**Advanced Features:**



|                               |         |
|-------------------------------|---------|
| Auto Baudrate                 | Disable |
| TX Pin Active Level Inversion | Disable |
| RX Pin Active Level Inversion | Disable |
| Data Inversion                | Disable |
| TX and RX Pins Swapping       | Disable |
| Overrun                       | Enable  |
| DMA on RX Error               | Enable  |
| MSB First                     | Disable |

### 3.34. USB

**Mode: Device\_Only**

#### 3.34.1. Parameter Settings:

##### **Basic Parameters:**

|                       |                     |
|-----------------------|---------------------|
| Speed                 | Full Speed 12MBit/s |
| Physical interface    | Internal Phy        |
| Signal start of frame | Disabled            |

##### **Power Parameters:**

|                       |          |
|-----------------------|----------|
| Low Power             | Disabled |
| Link Power Management | Disabled |
| Battery Charging      | Disabled |

##### **EndPoint Parameters:**

|                    |          |
|--------------------|----------|
| Bulk double buffer | Disabled |
| Iso single buffer  | Disabled |

**\* User modified value**

## 4. System Configuration

### 4.1. GPIO configuration

| IP       | Pin              | Signal           | GPIO mode                     | GPIO pull/up pull down      | Max Speed | User Label |
|----------|------------------|------------------|-------------------------------|-----------------------------|-----------|------------|
| ADC1     | PC2              | ADC1_INP12       | Analog mode                   | No pull-up and no pull-down | n/a       | CC1        |
|          | PC3              | ADC1_INP13       | Analog mode                   | No pull-up and no pull-down | n/a       | CC2        |
|          | PB1              | ADC1_INP5        | Analog mode                   | No pull-up and no pull-down | n/a       |            |
|          | PF11             | ADC1_INP2        | Analog mode                   | No pull-up and no pull-down | n/a       |            |
| DAC1     | PA4              | DAC1_OUT1        | Analog mode                   | No pull-up and no pull-down | n/a       |            |
|          | PA5              | DAC1_OUT2        | Analog mode                   | No pull-up and no pull-down | n/a       |            |
| DEBUG    | PA14(JTCK/SWCLK) | DEBUG_JTCK-SWCLK | n/a                           | n/a                         | n/a       |            |
|          | PA13(JTMS/SWDIO) | DEBUG_JTMS-SWDIO | n/a                           | n/a                         | n/a       |            |
| ETH      | PC1              | ETH_MDC          | Alternate Function Push Pull  | No pull-up and no pull-down | High      |            |
|          | PA1              | ETH_REF_CLK      | Alternate Function Push Pull  | No pull-up and no pull-down | High      |            |
|          | PC4              | ETH_RXD0         | Alternate Function Push Pull  | No pull-up and no pull-down | High      |            |
|          | PA2              | ETH_MDIO         | Alternate Function Push Pull  | No pull-up and no pull-down | High      |            |
|          | PC5              | ETH_RXD1         | Alternate Function Push Pull  | No pull-up and no pull-down | High      |            |
|          | PB12             | ETH_TXD0         | Alternate Function Push Pull  | No pull-up and no pull-down | High      |            |
|          | PA7              | ETH_CRS_DV       | Alternate Function Push Pull  | No pull-up and no pull-down | High      |            |
|          | PB11             | ETH_TX_EN        | Alternate Function Push Pull  | No pull-up and no pull-down | High      |            |
|          | PB15             | ETH_TXD1         | Alternate Function Push Pull  | No pull-up and no pull-down | High      |            |
| FDCAN1   | PE1              | FDCAN1_TX        | Alternate Function Push Pull  | No pull-up and no pull-down | Low       |            |
|          | PE0              | FDCAN1_RX        | Alternate Function Push Pull  | No pull-up and no pull-down | Low       |            |
| I2C2     | PF0              | I2C2_SDA         | Alternate Function Open Drain | No pull-up and no pull-down | Low       | MGMNT_SDA  |
|          | PF1              | I2C2_SCL         | Alternate Function Open Drain | No pull-up and no pull-down | Low       | MGMNT_SCL  |
| I2C4     | PF5              | I2C4_SCL         | Alternate Function Open Drain | No pull-up and no pull-down | Low       |            |
|          | PH12             | I2C4_SDA         | Alternate Function Open Drain | No pull-up and no pull-down | Low       |            |
| I3C1     | PB8              | I3C1_SCL         | Alternate Function Push Pull  | Pull-up                     | Very High |            |
|          | PB9              | I3C1_SDA         | Alternate Function Push Pull  | Pull-up                     | Very High |            |
| OCTOSPI1 | PF7              | OCTOSPI1_IO2     | Alternate Function Push Pull  | No pull-up and no pull-down | Very High |            |
|          | PF6              | OCTOSPI1_IO3     | Alternate Function Push Pull  | No pull-up and no pull-down | Very High |            |
|          | PF10             | OCTOSPI1_CLK     | Alternate Function Push Pull  | No pull-up and no pull-down | Very High |            |
|          | PF9              | OCTOSPI1_IO1     | Alternate Function Push Pull  | No pull-up and no pull-down | Very High |            |
|          | PF8              | OCTOSPI1_IO0     | Alternate Function Push Pull  | No pull-up and no pull-down | Very High |            |
|          | PE11             | OCTOSPI1_NCS     | Alternate Function Push Pull  | No pull-up and no pull-down | Very High |            |

| IP     | Pin                        | Signal        | GPIO mode                    | GPIO pull/up pull down      | Max Speed | User Label |
|--------|----------------------------|---------------|------------------------------|-----------------------------|-----------|------------|
| RCC    | PC14-OSC32_IN( OSC32_IN)   | RCC_OSC32_IN  | n/a                          | n/a                         | n/a       |            |
|        | PC15-OSC32_OUT( OSC32_OUT) | RCC_OSC32_OUT | n/a                          | n/a                         | n/a       |            |
|        | PC9                        | RCC_MCO_2     | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
|        | PH0-OSC_IN(PH0)            | RCC_OSC_IN    | n/a                          | n/a                         | n/a       |            |
|        | PH1-OSC_OUT(PH1)           | RCC_OSC_OUT   | n/a                          | n/a                         | n/a       |            |
| SDMMC2 | PD7                        | SDMMC2_CMD    | Alternate Function Push Pull | No pull-up and no pull-down | High      |            |
|        | PG12                       | SDMMC2_D3     | Alternate Function Push Pull | No pull-up and no pull-down | High      |            |
|        | PG11                       | SDMMC2_D2     | Alternate Function Push Pull | No pull-up and no pull-down | High      |            |
|        | PG10                       | SDMMC2_D1     | Alternate Function Push Pull | No pull-up and no pull-down | High      |            |
|        | PD6                        | SDMMC2_CK     | Alternate Function Push Pull | No pull-up and no pull-down | High      |            |
|        | PG9                        | SDMMC2_D0     | Alternate Function Push Pull | No pull-up and no pull-down | High      |            |
| SPI2   | PI3                        | SPI2_MOSI     | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
|        | PI2                        | SPI2_MISO     | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
|        | PI1                        | SPI2_SCK      | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
| SPI3   | PB4(NJTSTRST)              | SPI3_MISO     | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
|        | PC12                       | SPI3_MOSI     | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
|        | PC10                       | SPI3_SCK      | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
| TIM1   | PH9                        | TIM1_CH2      | Alternate Function Push Pull | No pull-up and no pull-down | Low       | LED_2      |
|        | PE13                       | TIM1_CH3      | Alternate Function Push Pull | No pull-up and no pull-down | Low       | LED_3      |
|        | PE9                        | TIM1_CH1      | Alternate Function Push Pull | No pull-up and no pull-down | Low       | LED_1      |
|        | PE14                       | TIM1_CH4      | Alternate Function Push Pull | No pull-up and no pull-down | Low       | LED_4      |
| TIM2   | PB3(JTDO/TRACESWO)         | TIM2_CH2      | Alternate Function Push Pull | No pull-up and no pull-down | Low       | LED_6      |
|        | PA15(JTDI)                 | TIM2_CH1      | Alternate Function Push Pull | No pull-up and no pull-down | Low       | LED_5      |
|        | PA3                        | TIM2_CH4      | Alternate Function Push Pull | No pull-up and no pull-down | Low       | LED_8      |
|        | PB10                       | TIM2_CH3      | Alternate Function Push Pull | No pull-up and no pull-down | Low       | LED_7      |
| TIM3   | PC6                        | TIM3_CH1      | Alternate Function Push Pull | No pull-up and no pull-down | Low       | PWM_FAN    |
| TIM13  | PA6                        | TIM13_CH1     | Alternate Function Push Pull | No pull-up and no pull-down | Low       | ARGB       |
| UART5  | PB5                        | UART5_RX      | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
|        | PB6                        | UART5_TX      | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
| UART9  | PD15                       | UART9_TX      | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
|        | PD14                       | UART9_RX      | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
| UCPD1  | PB13                       | UCPD1_CC1     | Analog mode                  | No pull-up and no pull-down | n/a       |            |
|        |                            |               |                              |                             |           |            |

| IP                    | Pin  | Signal      | GPIO mode                                                          | GPIO pull/up pull down      | Max Speed | User Label     |
|-----------------------|------|-------------|--------------------------------------------------------------------|-----------------------------|-----------|----------------|
|                       | PB14 | UCPD1_CC2   | Analog mode                                                        | No pull-up and no pull-down | n/a       |                |
| USART1                | PA10 | USART1_RX   | Alternate Function Push Pull                                       | No pull-up and no pull-down | Low       |                |
|                       | PA9  | USART1_TX   | Alternate Function Push Pull                                       | No pull-up and no pull-down | Low       |                |
| USART10               | PE3  | USART10_TX  | Alternate Function Push Pull                                       | No pull-up and no pull-down | Low       |                |
|                       | PE2  | USART10_RX  | Alternate Function Push Pull                                       | No pull-up and no pull-down | Low       |                |
|                       | PG14 | USART10_RTS | Alternate Function Push Pull                                       | No pull-up and no pull-down | Low       |                |
|                       | PG13 | USART10_CTS | Alternate Function Push Pull                                       | No pull-up and no pull-down | Low       |                |
| USART11               | PF3  | USART11_TX  | Alternate Function Push Pull                                       | No pull-up and no pull-down | Low       |                |
|                       | PF4  | USART11_RX  | Alternate Function Push Pull                                       | No pull-up and no pull-down | Low       |                |
| USB                   | PA12 | USB_DP      | n/a                                                                | n/a                         | n/a       |                |
|                       | PA11 | USB_DM      | n/a                                                                | n/a                         | n/a       |                |
| Single Mapped Signals | PH10 | I2C4_SMBA   | Alternate Function Open Drain                                      | No pull-up and no pull-down | Low       |                |
| GPIO                  | PE4  | GPIO_EXTI4  | <b>External Interrupt Mode with Falling edge trigger detection</b> | No pull-up and no pull-down | n/a       |                |
|                       | PI7  | GPIO_Output | Output Push Pull                                                   | No pull-up and no pull-down | Low       | BUCK_ENABLE    |
|                       | PI6  | GPIO_EXTI6  | <b>External Interrupt Mode with Falling edge trigger detection</b> | No pull-up and no pull-down | n/a       | BUCK_INT       |
|                       | PI5  | GPIO_Input  | Input mode                                                         | No pull-up and no pull-down | n/a       |                |
|                       | PC13 | GPIO_Input  | Input mode                                                         | No pull-up and no pull-down | n/a       | MODULE_PRESENT |
|                       | PI8  | GPIO_Output | Output Push Pull                                                   | No pull-up and no pull-down | Low       | GPIO_OD_1      |
|                       | PI9  | GPIO_Output | Output Push Pull                                                   | No pull-up and no pull-down | Low       |                |
|                       | PI4  | GPIO_Input  | Input mode                                                         | No pull-up and no pull-down | n/a       | GPIO_PP_2      |
|                       | PD3  | GPIO_Input  | Input mode                                                         | No pull-up and no pull-down | n/a       | USB_HS         |
|                       | PD2  | GPIO_Input  | Input mode                                                         | No pull-up and no pull-down | n/a       | USB_SUSP       |
|                       | PI11 | GPIO_Input  | Input mode                                                         | No pull-up and no pull-down | n/a       | GPIO_OD_IN1    |
|                       | PH13 | GPIO_Output | Output Push Pull                                                   | No pull-up and no pull-down | Low       | USB_RESET      |
|                       | PH2  | GPIO_Input  | Input mode                                                         | No pull-up and no pull-down | n/a       | GPIO_OD_IN2    |
|                       | PH4  | GPIO_Input  | Input mode                                                         | No pull-up and no pull-down | n/a       | DUT_PRESENT    |
|                       | PH5  | GPIO_Output | Output Push Pull                                                   | No pull-up and no pull-down | Low       | DUT_RESET      |
|                       | PG7  | GPIO_Input  | Input mode                                                         | No pull-up and no pull-down | n/a       | DBG_IN1        |
|                       | PG6  | GPIO_Input  | Input mode                                                         | No pull-up and no pull-down | n/a       | DBG_IN2        |
|                       | PG4  | GPIO_Output | Output Push Pull                                                   | No pull-up and no pull-down | Low       | DBG_LED1       |
|                       | PG3  | GPIO_Output | Output Push Pull                                                   | No pull-up and no pull-down | Low       | DBG_LED2       |
|                       | PG2  | GPIO_Output | Output Push Pull                                                   | No pull-up and no pull-down | Low       | INTERCOM_OUT   |
|                       | PG1  | GPIO_Output | Output Push Pull                                                   | No pull-up and no pull-down | Low       | DGB_O1         |
|                       | PH6  | GPIO_Input  | Input mode                                                         | No pull-up and no pull-down | n/a       | ACT0           |
|                       |      |             |                                                                    |                             |           |                |

| IP | Pin  | Signal      | GPIO mode                                                          | GPIO pull/up pull down      | Max Speed             | User Label  |
|----|------|-------------|--------------------------------------------------------------------|-----------------------------|-----------------------|-------------|
|    | PH8  | GPIO_Input  | Input mode                                                         | No pull-up and no pull-down | n/a                   | ACT1        |
|    | PD13 | GPIO_Input  | Input mode                                                         | No pull-up and no pull-down | n/a                   | INTERCOM_IN |
|    | PG0  | GPIO_Output | Output Push Pull                                                   | No pull-up and no pull-down | Low                   | DBG_O2      |
|    | PH7  | GPIO_Input  | Input mode                                                         | No pull-up and no pull-down | n/a                   | ACT2        |
|    | PD12 | GPIO_Input  | Input mode                                                         | No pull-up and no pull-down | n/a                   | REQ_RESET   |
|    | PD10 | EVENTOUT    | Alternate Function Push Pull                                       | No pull-up and no pull-down | <b>Very High</b><br>* |             |
|    | PF15 | GPIO_Output | Output Push Pull                                                   | No pull-up and no pull-down | Low                   | DBG_O3      |
|    | PD9  | GPIO_Output | Output Push Pull                                                   | No pull-up and no pull-down | Low                   | ETH_RESET   |
|    | PD8  | GPIO_EXTI8  | <b>External Interrupt Mode with Falling edge trigger detection</b> | No pull-up and no pull-down | n/a                   |             |
|    | PF14 | GPIO_Output | Output Push Pull                                                   | No pull-up and no pull-down | Low                   | DBG_O4      |

## 4.2. GPDMA1

## 4.3. GPDMA2

## 4.4. LINKEDLIST

## 4.5. NVIC configuration

### 4.5.1. NVIC

| Interrupt Table                         | Enable | Preenmption Priority | SubPriority |
|-----------------------------------------|--------|----------------------|-------------|
| Non maskable interrupt                  | true   | 0                    | 0           |
| Hard fault interrupt                    | true   | 0                    | 0           |
| Memory management fault                 | true   | 0                    | 0           |
| Pre-fetch fault, memory access fault    | true   | 0                    | 0           |
| Undefined instruction or illegal state  | true   | 0                    | 0           |
| System service call via SWI instruction | true   | 0                    | 0           |
| Debug monitor                           | true   | 0                    | 0           |
| Pendable request for system service     | true   | 0                    | 0           |
| System tick timer                       | true   | 15                   | 0           |
| TIM6 global interrupt                   | true   | 0                    | 0           |
| Flash non-secure global interrupt       | unused |                      |             |
| RCC non-secure global interrupt         | unused |                      |             |
| EXTI Line4 interrupt                    | unused |                      |             |
| EXTI Line6 interrupt                    | unused |                      |             |
| EXTI Line8 interrupt                    | unused |                      |             |
| ADC1 global interrupt                   | unused |                      |             |
| DAC1 interrupt                          | unused |                      |             |
| FDCAN1 interrupt 0                      | unused |                      |             |
| FDCAN1 interrupt 1                      | unused |                      |             |
| TIM1 Break interrupt                    | unused |                      |             |
| TIM1 Update interrupt                   | unused |                      |             |
| TIM1 Trigger and Commutation interrupts | unused |                      |             |
| TIM1 Capture Compare interrupt          | unused |                      |             |
| TIM2 global interrupt                   | unused |                      |             |
| TIM3 global interrupt                   | unused |                      |             |
| I2C2 Event interrupt                    | unused |                      |             |
| I2C2 Error interrupt                    | unused |                      |             |
| SPI2 global interrupt                   | unused |                      |             |
| SPI3 global interrupt                   | unused |                      |             |
| USART1 global interrupt                 | unused |                      |             |
| UART5 global interrupt                  | unused |                      |             |
| USB FS global interrupt                 | unused |                      |             |
| UCPD1 global interrupt                  | unused |                      |             |
| OCTOSPI1 global interrupt               | unused |                      |             |
| USART10 global interrupt                | unused |                      |             |
| USART11 global interrupt                | unused |                      |             |
| UART9 global interrupt                  | unused |                      |             |
| SDMMC2 global interrupt                 | unused |                      |             |

| Interrupt Table                    | Enable | Preenmption Priority | SubPriority |
|------------------------------------|--------|----------------------|-------------|
| FPU global interrupt               |        | unused               |             |
| Instruction cache global interrupt |        | unused               |             |
| Data cache global interrupt        |        | unused               |             |
| Ethernet global interrupt          |        | unused               |             |
| Ethernet Wakeup global interrupt   |        | unused               |             |
| CORDIC global interrupt            |        | unused               |             |
| FMAC global interrupt              |        | unused               |             |
| TIM13 global interrupt             |        | unused               |             |
| I3C1 Event interrupt               |        | unused               |             |
| I3C1 Error interrupt               |        | unused               |             |
| I2C4 Event interrupt               |        | unused               |             |
| I2C4 Error interrupt               |        | unused               |             |

#### 4.5.2. NVIC Code generation

| Enabled interrupt Table                 | Select for init sequence ordering | Generate IRQ handler | Call HAL handler |
|-----------------------------------------|-----------------------------------|----------------------|------------------|
| Non maskable interrupt                  | false                             | true                 | false            |
| Hard fault interrupt                    | false                             | true                 | false            |
| Memory management fault                 | false                             | true                 | false            |
| Pre-fetch fault, memory access fault    | false                             | true                 | false            |
| Undefined instruction or illegal state  | false                             | true                 | false            |
| System service call via SWI instruction | false                             | true                 | false            |
| Debug monitor                           | false                             | true                 | false            |
| Pendable request for system service     | false                             | true                 | false            |
| System tick timer                       | false                             | true                 | true             |
| TIM6 global interrupt                   | false                             | true                 | true             |

\* User modified value

## 5. System Views

### 5.1. Category view

#### 5.1.1. Current

| Middleware  |        |         |              |            |          |           |                 |                   |            |       |
|-------------|--------|---------|--------------|------------|----------|-----------|-----------------|-------------------|------------|-------|
| System Core | Analog | Timers  | Connectivity | Multimedia | Security | Computing | Trace and Debug | Power and Thermal | Utilities  | Other |
| CORTEX_M33  | ADC1 ✓ | RTC ✓   | ETH ✓        | FDCAH1 ✓   |          | CORDIC ✓  | DEBUG ✓         | PWR ✓             | LINKEDLIST |       |
| DCACHE1 ✓   | DAC1 ✓ | TIM1 ✓  | I2C2 ✓       | I2C4 ✓     |          | CRC ✓     |                 |                   |            |       |
| GPDMA1      |        | TIM2 ✓  | I3C1 ✓       | OCTOSPH ✓  |          | FMAC ✓    |                 |                   |            |       |
| GPDMA2      |        | TIM3 ✓  | SDMMC2 ✓     | SPI2 ✓     |          |           |                 |                   |            |       |
| GPIO ⚠      |        | TIM13 ✓ | SPI3 ✓       | UART5 ✓    |          |           |                 |                   |            |       |
| ICACHE ✓    |        |         | UART9 ✓      | UCPD1 ✓    |          |           |                 |                   |            |       |
| NVIC ✓      |        |         | USART1 ✓     | USART10 ✓  |          |           |                 |                   |            |       |
| RCC ✓       |        |         | USART11 ✓    | USB ✓      |          |           |                 |                   |            |       |
| SYS ✓       |        |         |              |            |          |           |                 |                   |            |       |



## 6. Docs & Resources

| Type                    | Link                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BSDL files              | <a href="https://www.st.com/resource/en/bsdl_model/stm32h5-bsdl.zip">https://www.st.com/resource/en/bsdl_model/stm32h5-bsdl.zip</a>                                                                                                                                                                                                                                                                                     |
| IBIS models             | <a href="https://www.st.com/resource/en/ibis_model/stm32h5-ibis.zip">https://www.st.com/resource/en/ibis_model/stm32h5-ibis.zip</a>                                                                                                                                                                                                                                                                                     |
| System View Description | <a href="https://www.st.com/resource/en/svd/stm32h5-svd.zip">https://www.st.com/resource/en/svd/stm32h5-svd.zip</a>                                                                                                                                                                                                                                                                                                     |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf">https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf</a>                                                                                                                                                                                                             |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf">https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf</a>                                                                                                                                                                                                                                     |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf">https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf</a>                                                                                                                                                                                                               |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h5-series-overview.pdf">https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h5-series-overview.pdf</a>                                                                                                                                                                                                         |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/faq-security-stm32h5.pdf">https://www.st.com/resource/en/product_presentation/faq-security-stm32h5.pdf</a>                                                                                                                                                                                                                                                 |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf">https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf</a>                                                                                                                                                                                                             |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/secure-manager-introduction-v1.pdf">https://www.st.com/resource/en/product_presentation/secure-manager-introduction-v1.pdf</a>                                                                                                                                                                                                                             |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-entry-level-graphics.pdf">https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-entry-level-graphics.pdf</a>                                                                                                                                                                                                   |
| Brochures               | <a href="https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf">https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf</a>                                                                                                                                                                                                                     |
| Brochures               | <a href="https://www.st.com/resource/en/brochure/expansion-boards-for-intelligent-power-switches.pdf">https://www.st.com/resource/en/brochure/expansion-boards-for-intelligent-power-switches.pdf</a>                                                                                                                                                                                                                   |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flstm32nucleo.pdf">https://www.st.com/resource/en/flyer/flstm32nucleo.pdf</a>                                                                                                                                                                                                                                                                                             |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flstm32h5.pdf">https://www.st.com/resource/en/flyer/flstm32h5.pdf</a>                                                                                                                                                                                                                                                                                                     |
| Security Bulletin       | <a href="https://www.st.com/resource/en/technical_note/tn1474-security-bulletin-tn1474stpsirt-information-on-softwarebased--microarchitectural-timing-sidechannel-attacks-on-mcus-with-trustzone-for--armv8m-">https://www.st.com/resource/en/technical_note/tn1474-security-bulletin-tn1474stpsirt-information-on-softwarebased--microarchitectural-timing-sidechannel-attacks-on-mcus-with-trustzone-for--armv8m-</a> |

|                   |                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | stmicroelectronics.pdf                                                                                                                                                                                                                                                                                                                                                                      |
| Security Bulletin | <a href="https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf</a>                                                     |
| Security Bulletin | <a href="https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-protection-statement-for-stmicroelectronics-certified-products-stmicroelectronics.pdf">https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-protection-statement-for-stmicroelectronics-certified-products-stmicroelectronics.pdf</a>                                                                   |
| Application Notes | <a href="https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf</a>                                                                                                                   |
| Application Notes | <a href="https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf</a>                                                                                       |
| Application Notes | <a href="https://www.st.com/resource/en/application_note/an3155-uart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an3155-uart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf</a>                                                                                                                             |
| Application Notes | <a href="https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf</a>                                                                                                                       |
| Application Notes | <a href="https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf</a>                                                                                                                               |
| Application Notes | <a href="https://www.st.com/resource/en/application_note/an4286-spi-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an4286-spi-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf</a>                                                                                                                               |
| Application Notes | <a href="https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf</a>                               |
| Application Notes | <a href="https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf</a>                                                                                                                       |
| Application Notes | <a href="https://www.st.com/resource/en/application_note/an4776-generalpurpose-timer-cookbook-for-stm32-microcontrollers-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an4776-generalpurpose-timer-cookbook-for-stm32-microcontrollers-stmicroelectronics.pdf</a>                                                                                                 |
| Application Notes | <a href="https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf</a> |
| Application Notes | <a href="https://www.st.com/resource/en/application_note/an4989-stm32-microcontroller-debug-toolbox-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an4989-stm32-microcontroller-debug-toolbox-stmicroelectronics.pdf</a>                                                                                                                                           |
| Application Notes | <a href="https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-">https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-</a>                                                                                                                                                                                                                               |

digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes [https://www.st.com/resource/en/application\\_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an4991-how-to-wake-up-an-stm32-microcontroller-from-lowpower-mode-with-the-usart-or-the-lpuart-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4991-how-to-wake-up-an-stm32-microcontroller-from-lowpower-mode-with-the-usart-or-the-lpuart-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an5325-how-to-use-the-cordic-to-perform-mathematical-functions-on-stm32-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5325-how-to-use-the-cordic-to-perform-mathematical-functions-on-stm32-mcus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an5925-stm32cube-mcu-package-examples-for-stm32h5-series-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5925-stm32cube-mcu-package-examples-for-stm32h5-series-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an5927-i3c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5927-i3c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an5930-guidelines-for-power-management-on-stm32h5-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5930-guidelines-for-power-management-on-stm32h5-mcus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an5711-getting-started-with-stm32h5-mcu-hardware-development-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5711-getting-started-with-stm32h5-mcu-hardware-development-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an5688-migrating-microcontroller-applications-from-stm32f427437-and-stm32f429439-to-stm32h573563-and-stm32h562-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5688-migrating-microcontroller-applications-from-stm32f427437-and-stm32f429439-to-stm32h573563-and-stm32h562-mcus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an4879-introduction-to-usb-hardware-and-pcb-guidelines-using-stm32-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4879-introduction-to-usb-hardware-and-pcb-guidelines-using-stm32-mcus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an5225-introduction-to-usb-typec-power-delivery-for-stm32-mcus-and-mpus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5225-introduction-to-usb-typec-power-delivery-for-stm32-mcus-and-mpus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an5342--how-to-use-](https://www.st.com/resource/en/application_note/an5342--how-to-use-)

error-correction-code-ecc-management-for-internal-memories-protection-on-stm32-mcus-stmicroelectronics.pdf

Application Notes [https://www.st.com/resource/en/application\\_note/an5872-introduction-to-the-system-architecture-and-performance-in-the-stm32h5-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5872-introduction-to-the-system-architecture-and-performance-in-the-stm32h5-mcus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an4894-how-to-use-eeeprom-emulation-on-stm32-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4894-how-to-use-eeeprom-emulation-on-stm32-mcus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an5537-how-to-use-adc-oversampling-techniques-to-improve-signal-to-noise-ratio-on-stm32-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5537-how-to-use-adc-oversampling-techniques-to-improve-signal-to-noise-ratio-on-stm32-mcus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an5036-guidelines-for-thermal-management-on-stm32-applications-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5036-guidelines-for-thermal-management-on-stm32-applications-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an4992-introduction-to-secure-firmware-install-sfi-for-stm32-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4992-introduction-to-secure-firmware-install-sfi-for-stm32-mcus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an5405-how-to-use-fdcan-bootloader-protocol-on-stm32-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5405-how-to-use-fdcan-bootloader-protocol-on-stm32-mcus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an6014-introduction-to-sfspi-versions-for-stm32h5-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an6014-introduction-to-sfspi-versions-for-stm32h5-mcus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an5690-how-to-use-vrefbuf-peripheral-on-stm32-mcus-and-mpus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5690-how-to-use-vrefbuf-peripheral-on-stm32-mcus-and-mpus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an4230-introduction-to-random-number-generation-validation-using-the-nist-statistical-test-suite-for-stm32-mcus-and-mpus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4230-introduction-to-random-number-generation-validation-using-the-nist-statistical-test-suite-for-stm32-mcus-and-mpus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an2867-guidelines-for-oscillator-design-on-stm8afals-and-stm32-mcus-mpus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an2867-guidelines-for-oscillator-design-on-stm8afals-and-stm32-mcus-mpus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an4013-introduction-to-timers-for-stm32-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4013-introduction-to-timers-for-stm32-mcus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an4277-how-to-use-pwm-shutdown-for-motor-control-and-digital-power-conversion-on-stm32-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4277-how-to-use-pwm-shutdown-for-motor-control-and-digital-power-conversion-on-stm32-mcus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an4759-introduction-to-using-the-hardware-realtime-clock-rtc-and-the-tamper-management-unit](https://www.st.com/resource/en/application_note/an4759-introduction-to-using-the-hardware-realtime-clock-rtc-and-the-tamper-management-unit)

tamp-with-stm32-mcus-stmicroelectronics.pdf

Application Notes [https://www.st.com/resource/en/application\\_note/an4908-getting-started-with-uart-automatic-baud-rater-detection-for-stm32-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4908-getting-started-with-uart-automatic-baud-rater-detection-for-stm32-mcus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an5156-introduction-to-security-for-stm32-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5156-introduction-to-security-for-stm32-mcus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an5212-how-to-use-stm32-cache-to-optimize-performance-and-power-efficiency-for-stm32-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5212-how-to-use-stm32-cache-to-optimize-performance-and-power-efficiency-for-stm32-mcus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an5593-how-to-use-the-gpdma-for-stm32-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5593-how-to-use-the-gpdma-for-stm32-mcus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an5879-introduction-to-i3c-for-stm32-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5879-introduction-to-i3c-for-stm32-mcus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an5543-guidelines-for-enhanced-spi-communication-on-stm32-mcus-and-mpus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5543-guidelines-for-enhanced-spi-communication-on-stm32-mcus-and-mpus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an5959-migrating-within-the-stm32h5-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5959-migrating-within-the-stm32h5-mcus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an5050-getting-started-with-octospi-hexadecapi-and-xsapi-interface-on-stm32-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5050-getting-started-with-octospi-hexadecapi-and-xsapi-interface-on-stm32-mcus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an6008-getting-started-with-debug-authentication-da-for-stm32-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an6008-getting-started-with-debug-authentication-da-for-stm32-mcus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an5348-introduction-to-fdcan-peripherals-for-stm32-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5348-introduction-to-fdcan-peripherals-for-stm32-mcus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/cd00211314-how-to-optimize-the-adc-accuracy-in-the-stm32-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/cd00211314-how-to-optimize-the-adc-accuracy-in-the-stm32-mcus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an2639-soldering-recommendations-and-package-information-for-leadfree-ecopack2-mcus-and-mpus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an2639-soldering-recommendations-and-package-information-for-leadfree-ecopack2-mcus-and-mpus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an5020-introduction-to-digital-camera-interface-dcml-for-stm32-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5020-introduction-to-digital-camera-interface-dcml-for-stm32-mcus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an6202-how-to-](https://www.st.com/resource/en/application_note/an6202-how-to-)

calibrate-internal-rc-oscillators-on-stm32h5-mcus-stmicroelectronics.pdf

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| Application Notes                                    | <a href="https://www.st.com/resource/en/application_note/an6246-guidelines-for-estimating-stm32h5-mcus-lifetime-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an6246-guidelines-for-estimating-stm32h5-mcus-lifetime-stmicroelectronics.pdf</a>                                                                                                                       |
| Application Notes                                    | <a href="https://www.st.com/resource/en/application_note/an4566-how-to-extend-the-dac-performance-on-stm32-mcus-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an4566-how-to-extend-the-dac-performance-on-stm32-mcus-stmicroelectronics.pdf</a>                                                                                                                       |
| Application Notes                                    | <a href="https://www.st.com/resource/en/application_note/an5647-the-goertzel-algorithm-to-compute-individual-terms-of-the-discrete-fourier-transform-dft-in-stm32-products-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an5647-the-goertzel-algorithm-to-compute-individual-terms-of-the-discrete-fourier-transform-dft-in-stm32-products-stmicroelectronics.pdf</a> |
| Application Notes                                    | <a href="https://www.st.com/resource/en/application_note/an2606-introduction-to-system-memory-boot-mode-on-stm32-mcus-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an2606-introduction-to-system-memory-boot-mode-on-stm32-mcus-stmicroelectronics.pdf</a>                                                                                                           |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an4657-stm32-inapplication-programming-iap-using-the-usart-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an4657-stm32-inapplication-programming-iap-using-the-usart-stmicroelectronics.pdf</a>                                                                                                               |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an4841-digital-signal-processing-for-stm32-microcontrollers-using-cmsis-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an4841-digital-signal-processing-for-stm32-microcontrollers-using-cmsis-stmicroelectronics.pdf</a>                                                                                     |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an5360-getting-started-with-projects-based-on-the-stm32mp1-series-in-stm32cubeide-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an5360-getting-started-with-projects-based-on-the-stm32mp1-series-in-stm32cubeide-stmicroelectronics.pdf</a>                                                                 |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an5361-getting-started-with-projects-based-on-dualcore-stm32h7-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an5361-getting-started-with-projects-based-on-dualcore-stm32h7-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf</a>                                     |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an5394-getting-started-with-projects-based-on-the-stm32l5-series-in-stm32cubeide-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an5394-getting-started-with-projects-based-on-the-stm32l5-series-in-stm32cubeide-stmicroelectronics.pdf</a>                                                                   |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an5418-how-to-build-a-simple-usbp-d-sink-application-with-stm32cubemx-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an5418-how-to-build-a-simple-usbp-d-sink-application-with-stm32cubemx-stmicroelectronics.pdf</a>                                                                                         |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an5426-migrating-graphics-middleware-projects-from-stm32cubemx-540-to-stm32cubemx-550-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an5426-migrating-graphics-middleware-projects-from-stm32cubemx-540-to-stm32cubemx-550-stmicroelectronics.pdf</a>                                                         |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an5564-getting-started-with-projects-based-on-dualcore-stm32wl-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an5564-getting-started-with-projects-based-on-dualcore-stm32wl-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf</a>                                     |

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| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an5731-stm32cubemx-and-stm32cubeide-threadsafe-solution-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an5731-stm32cubemx-and-stm32cubeide-threadsafe-solution-stmicroelectronics.pdf</a>                                               |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an5925-stm32cube-mcu-package-examples-for-stm32h5-series-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an5925-stm32cube-mcu-package-examples-for-stm32h5-series-stmicroelectronics.pdf</a>                                             |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an4502-stm32-smbuspmibus-expansion-package-for-stm32cube-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an4502-stm32-smbuspmibus-expansion-package-for-stm32cube-stmicroelectronics.pdf</a>                                             |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an4879-introduction-to-usb-hardware-and-pcb-guidelines-using-stm32-mcus-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an4879-introduction-to-usb-hardware-and-pcb-guidelines-using-stm32-mcus-stmicroelectronics.pdf</a>               |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an5952-how-to-use-cmake-in-stm32cubeide-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an5952-how-to-use-cmake-in-stm32cubeide-stmicroelectronics.pdf</a>                                                                               |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an6007-getting-started-with-stirot-st-immutable-root-of-trust-for-stm32h5-mcus-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an6007-getting-started-with-stirot-st-immutable-root-of-trust-for-stm32h5-mcus-stmicroelectronics.pdf</a> |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an5054-how-to-perform-secure-programming-using-stm32cubeprogrammer-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an5054-how-to-perform-secure-programming-using-stm32cubeprogrammer-stmicroelectronics.pdf</a>                         |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an6127-getting-started-with-stm32h7rx7sx-mcus-in-stm32cubeide-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an6127-getting-started-with-stm32h7rx7sx-mcus-in-stm32cubeide-stmicroelectronics.pdf</a>                                   |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an6265-getting-started-with-stm32n6-mcus-in-stm32cubeide-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an6265-getting-started-with-stm32n6-mcus-in-stm32cubeide-stmicroelectronics.pdf</a>                                             |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an6382-introduction-to-swd-multidrop-for-stm32-with-stlink-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an6382-introduction-to-swd-multidrop-for-stm32-with-stlink-stmicroelectronics.pdf</a>                                         |
| Errata Sheets                                        | <a href="https://www.st.com/resource/en/errata_sheet/es0565-stm32h562xx563xx573xx-device-errata-stmicroelectronics.pdf">https://www.st.com/resource/en/errata_sheet/es0565-stm32h562xx563xx573xx-device-errata-stmicroelectronics.pdf</a>                                                                                 |
| Datasheet                                            | <a href="https://www.st.com/resource/en/datasheet/dm00956662.pdf">https://www.st.com/resource/en/datasheet/dm00956662.pdf</a>                                                                                                                                                                                             |
| Programming                                          | <a href="https://www.st.com/resource/en/programming_manual/pm0264-stm32-">https://www.st.com/resource/en/programming_manual/pm0264-stm32-</a>                                                                                                                                                                             |

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| Manuals                       | <a href="#">cortexm33-mcus-and-mpus-programming-manual-stmicroelectronics.pdf</a>                                                                                                                                                                                                                                                                             |
| Reference<br>Manuals          | <a href="https://www.st.com/resource/en/reference_manual/rm0481-stm32h52333xx-stm32h56263xx-and-stm32h573xx-armbased-32bit-mcus-stmicroelectronics.pdf">https://www.st.com/resource/en/reference_manual/rm0481-stm32h52333xx-stm32h56263xx-and-stm32h573xx-armbased-32bit-mcus-stmicroelectronics.pdf</a>                                                     |
| Technical Notes<br>& Articles | <a href="https://www.st.com/resource/en/technical_note/tn1163-description-of-wlcsp-for-microcontrollers-and-recommendations-for-its-use-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1163-description-of-wlcsp-for-microcontrollers-and-recommendations-for-its-use-stmicroelectronics.pdf</a>                                     |
| Technical Notes<br>& Articles | <a href="https://www.st.com/resource/en/technical_note/tn1204-tape-and-reel-shipping-media-for-stm32-microcontrollers-in-bga-packages-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1204-tape-and-reel-shipping-media-for-stm32-microcontrollers-in-bga-packages-stmicroelectronics.pdf</a>                                         |
| Technical Notes<br>& Articles | <a href="https://www.st.com/resource/en/technical_note/tn1205-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-fpn-packages-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1205-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-fpn-packages-stmicroelectronics.pdf</a>                       |
| Technical Notes<br>& Articles | <a href="https://www.st.com/resource/en/technical_note/tn1206-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-qfp-packages-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1206-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-qfp-packages-stmicroelectronics.pdf</a>                       |
| Technical Notes<br>& Articles | <a href="https://www.st.com/resource/en/technical_note/tn1207-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-so-packages-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1207-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-so-packages-stmicroelectronics.pdf</a>                         |
| Technical Notes<br>& Articles | <a href="https://www.st.com/resource/en/technical_note/tn1208-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-tssop-and-ssop-packages-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1208-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-tssop-and-ssop-packages-stmicroelectronics.pdf</a> |
| Technical Notes<br>& Articles | <a href="https://www.st.com/resource/en/technical_note/tn1433-reference-device-marking-schematics-for-stm32-microcontrollers-and-microprocessors-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1433-reference-device-marking-schematics-for-stm32-microcontrollers-and-microprocessors-stmicroelectronics.pdf</a>                   |
| User Manuals                  | <a href="https://www.st.com/resource/en/user_manual/um3150-stm32h5-series-safety-manual-stmicroelectronics.pdf">https://www.st.com/resource/en/user_manual/um3150-stm32h5-series-safety-manual-stmicroelectronics.pdf</a>                                                                                                                                     |
| User Manuals                  | <a href="https://www.st.com/resource/en/user_manual/um3267-stm32h5-series-ulcsaiec-607301603351-selftest-library-user-guide-stmicroelectronics.pdf">https://www.st.com/resource/en/user_manual/um3267-stm32h5-series-ulcsaiec-607301603351-selftest-library-user-guide-stmicroelectronics.pdf</a>                                                             |